Monitoring data quality with Databricks Lakehouse Monitoring

Use Databricks Lakehouse Monitoring to proactively detect and respond to any deviations in your data distribution. Over time, your data may undergo changes in its underlying patterns. This could be feature drift, where the distribution of feature data changes over time, or concept drift, where the relationship between inputs and outputs of your model changes. Both types of drift can cause model quality to suffer. These changes can occur slowly or rapidly in your production environment, which is why monitoring your data even before it becomes an input into your ML models and data products is essential.

Mechanics of Lakehouse Monitoring

To monitor a table in Databricks, you create a monitor attached to that table. To monitor the performance of a ML model, you attach the monitor to an inference table that holds the model's inputs and corresponding predictions. Databricks Lakehouse Monitoring provides the following profile types of analysis: snapshot, time series, and inference.

In addition to selecting the table to be monitored, called the **primary table**, you can optionally specify a baseline table to reference for measuring drift or the change in values over time. A baseline table is useful when you have a sample of what you expect your data to look like, such as the data with which your model was trained. Lakehouse Monitoring automatically computes drift relative to expected data values and distributions of the baseline table.

Creating a table monitor automatically creates two metric tables, `profile_metrics` and `drift_metrics`. Lakehouse Monitoring computes metric values on the table for the time windows and data subsets or "slices" you specify when you create the monitor. You can also add your own custom metrics; see *Further reading* for details.

Visualization and alerting

Lakehouse Monitoring generates an SQL dashboard automatically for every monitor. These dashboards provide a crucial platform for examining metrics and acting on results. Databricks' alert system serves as a vigilant guardian, promptly notifying you of the significant shifts in data quality or distributions you subscribe to. *Figure 4.3* shows how all the Lakehouse Monitoring components work together, using the data from the primary table and optional baseline table to generate a profile metrics table and drift table, which then populate the dashboard and power alerts.